Scale Exercises - Section III, Level 22

Natural Minor Two-Octave Six-String Neck Ladder: A to A

The minor section still starts on A.

Extend the exact Level 21 formula across all six strings before moving it one fret.

1. Extend the Level 21 shape

Low E/A: R, R+2, R+3 (1,3,4). D: R, R+2 (1,3). G: R-1, R, R+2 (1,2,4). B: R, R+1, R+3 (1,2,4). High e: R (1).

2. Manage the two upper-string changes

The ascent uses 3-3-2-3-3-1 notes per string. Reach back one fret on G, then rebuild for the B-string tuning offset. Do not hold a static five-fret stretch.

3. Preserve rhythm and pick cycle

Repeat the high root at the turn. The complete up/down cell contains an even 30 notes, so every new key begins with the same pick orientation.

Worked example - A natural minor at low-E fret 5

Scale notes: A B C D E F G | A B C D E F G A. The high root is repeated at the turn.

LOW ROOT -> MIDDLE ROOT -> HIGH ROOT | HIGH ROOT -> MIDDLE ROOT -> LOW ROOT

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Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------5--5-----------------------------------------------|
B  |--------------------------------5--6--8--------8--6--5--------------------------------------|
G  |------------------------4--5--7-------------------------7--5--4-------------------------|
D  |-----------------5--7-----------------------------------------------7--5-------------------|
A  |---------5--7--8-------------------------------------------------------8--7--5-----------|
E  |5--7--8-----------------------------------------------------------------------8--7--5--|
```

KEY ROUTE A -> Bb -> B -> C -> C# -> D -> Eb -> E -> F -> F# -> G -> G# -> A, then reverse.

SPAN AUDIT The map spans R-1 to R+3, but the hand releases and reforms; never clamp a five-fret grip.

The fixed shape transposes. A natural minor is the starting rung, not the key of every rung.

Session protocol

1. Master the A-minor two-octave cell before adding a shift.
2. Start 10-15 BPM below your clean Level 21 tempo.
3. Use 2-3 neighboring rungs and pause one beat after each cell.
4. Repeat with a downstroke start and an upstroke start.
5. Remove the pause; connect the route and reverse at fret 17.
6. Add 2 BPM after 3 clean trips; reduce 4-6 BPM after 2 errors.

Level 22 - Chromatic Natural-Minor Route

Ascending route: complete each 30-note scale cell, then move the frame +1 fret Descending route: use the same cards in reverse order and move -1 fret

Rung	Key	Low-E root	Frame	Scale tones (degrees 1-7)
01	**A natural minor**	**fret 5**	**4-8**	**A B C D E F G**
02	Bb natural minor	fret 6	5-9	Bb C Db Eb F Gb Ab
03	B natural minor	fret 7	6-10	B C# D E F# G A
04	C natural minor	fret 8	7-11	C D Eb F G Ab Bb
05	C# natural minor	fret 9	8-12	C# D# E F# G# A B
06	D natural minor	fret 10	9-13	D E F G A Bb C
07	Eb natural minor	fret 11	10-14	Eb F Gb Ab Bb Cb Db
08	E natural minor	fret 12	11-15	E F# G A B C D
09	F natural minor	fret 13	12-16	F G Ab Bb C Db Eb
10	F# natural minor	fret 14	13-17	F# G# A B C# D E
11	G natural minor	fret 15	14-18	G A Bb C D Eb F
12	G# natural minor	fret 16	15-19	G# A# B C# D# E F#
13	**A natural minor - octave**	**fret 17**	**16-20**	**A B C D E F G**

How to use the map

Up pass:
rungs 01-13.

Turn:
finish the high-A cell at fret 17, then reverse.

Down pass:
rungs 12-01. Skip the high-A card unless starting a new round trip.

Span rule:
R-1 to R+3 is a moving frame, not a held five-fret stretch.

Goal:
hear the same minor formula while both octaves rise and fall.

Mastery gate before the next natural-minor level

Name every key before playing it; sing or say 1-2-b3-4-5-b6-b7; keep the G-string backreach relaxed; rebuild the B-string fingering instead of copying G; land each one-fret relocation without sliding; preserve equal note length and volume through all six strings; and complete the full fret-5 to fret-17 to fret-5 route with both pick starts.

ROUND-TRIP COUNT 25 cells x 30 notes = 750 picked notes.

Level 22 - Natural Minor Two-Octave Six-String Neck Ladder

Rungs 01-03: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

01 A MINOR

ROOT E5 | FRAME 4-8

Notes: A B C D E F G | A B C D E F G A

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------5--5-----------------------------------------|
B  |--------------------------------5--6--8-------8--6--5--------------------------------|
G  |------------------------4--5--7-------------------------7--5--4------------------------|
D  |-----------------5--7-----------------------------------------7--5-------------------|
A  |---------5--7--8-----------------------------------------------8--7--5----------|
E  |5--7--8-------------------------------------------------------------------8--7--5--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

02 BB MINOR

ROOT E6 | FRAME 5-9

Notes: Bb C Db Eb F Gb Ab | Bb C Db Eb F Gb Ab Bb

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------6--6-----------------------------------------|
B  |--------------------------------6--7--9-------9--7--6--------------------------------|
G  |------------------------5--6--8-------------------------8--6--5------------------------|
D  |-----------------6--8-----------------------------------------8--6-------------------|
A  |---------6--8--9-----------------------------------------------9--8--6----------|
E  |6--8--9-------------------------------------------------------------------9--8--6--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

03 B MINOR

ROOT E7 | FRAME 6-10

Notes: B C# D E F# G A | B C# D E F# G A B

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------7--7-----------------------------------------|
B  |--------------------------------7--8--10-------10-8--7--------------------------------|
G  |------------------------6--7--9-------------------------9--7--6------------------------|
D  |-----------------7--9-----------------------------------------9--7-------------------|
A  |---------7--9--10----------------------------------------------10-9--7----------|
E  |7--9--10------------------------------------------------------------------10-9--7--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 22 - Natural Minor Two-Octave Six-String Neck Ladder

Rungs 04-06: play each card as one cell, then move +1 fret

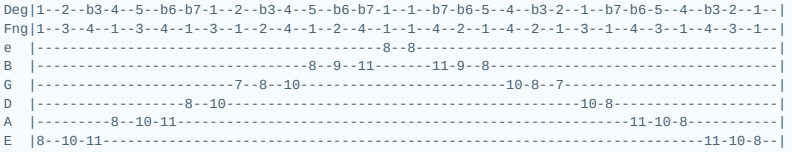
For the descending pass, read the cards in reverse order and move -1 fret

04 C MINOR

ROOT E8 | FRAME 7-11

Notes: C D Eb F G Ab Bb | C D Eb F G Ab Bb C

30-NOTE TWO-OCTAVE CELL



Return to the low root; release pressure; move the entire frame one fret.

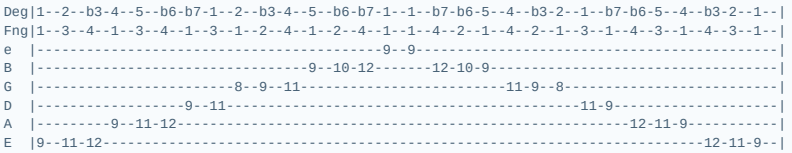
D-start and U-start

05 C# MINOR

ROOT E9 | FRAME 8-12

Notes: C# D# E F# G# A B | C# D# E F# G# A B C#

30-NOTE TWO-OCTAVE CELL



Return to the low root; release pressure; move the entire frame one fret.

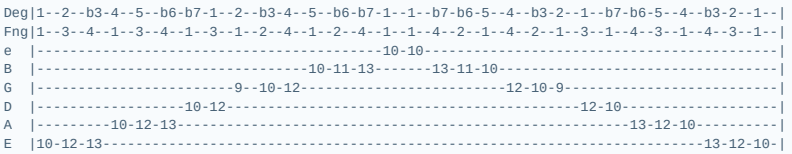
D-start and U-start

06 D MINOR

ROOT E10 | FRAME 9-13

Notes: D E F G A Bb C | D E F G A Bb C D

30-NOTE TWO-OCTAVE CELL



Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 22 - Natural Minor Two-Octave Six-String Neck Ladder

Rungs 07-09: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

07 EB MINOR

ROOT E11 | FRAME 10-14

Notes: Eb F Gb Ab Bb Cb Db | Eb F Gb Ab Bb Cb Db Eb

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------11-11-----------------------------------------|
B  |--------------------------------11-12-14-------14-12-11--------------------------------|
G  |------------------------10-11-13-------------------------13-11-10------------------------|
D  |-----------------11-13-----------------------------------------13-11------------------|
A  |---------11-13-14-------------------------------------------------14-13-11----------|
E  |11-13-14-------------------------------------------------------------------14-13-11-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

08 E MINOR

ROOT E12 | FRAME 11-15

Notes: E F# G A B C D | E F# G A B C D E

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------12-12-----------------------------------------|
B  |--------------------------------12-13-15-------15-13-12--------------------------------|
G  |------------------------11-12-14-------------------------14-12-11------------------------|
D  |-----------------12-14-----------------------------------------14-12------------------|
A  |---------12-14-15-------------------------------------------------15-14-12----------|
E  |12-14-15-------------------------------------------------------------------15-14-12-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

09 F MINOR

ROOT E13 | FRAME 12-16

Notes: F G Ab Bb C Db Eb | F G Ab Bb C Db Eb F

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------13-13-----------------------------------------|
B  |--------------------------------13-14-16-------16-14-13--------------------------------|
G  |------------------------12-13-15-------------------------15-13-12------------------------|
D  |-----------------13-15-----------------------------------------15-13------------------|
A  |---------13-15-16-------------------------------------------------16-15-13----------|
E  |13-15-16-------------------------------------------------------------------16-15-13-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 22 - Natural Minor Two-Octave Six-String Neck Ladder

Rungs 10-12: play each card as one cell, then move +1 fret

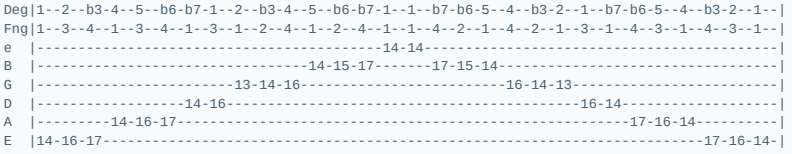
For the descending pass, read the cards in reverse order and move -1 fret

10 F# MINOR

ROOT E14 | FRAME 13-17

Notes: F# G# A B C# D E | F# G# A B C# D E F#

30-NOTE TWO-OCTAVE CELL



Return to the low root; release pressure; move the entire frame one fret.

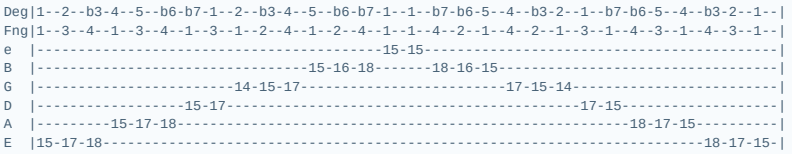
D-start and U-start

11 G MINOR

ROOT E15 | FRAME 14-18

Notes: G A Bb C D Eb F | G A Bb C D Eb F G

30-NOTE TWO-OCTAVE CELL



Return to the low root; release pressure; move the entire frame one fret.

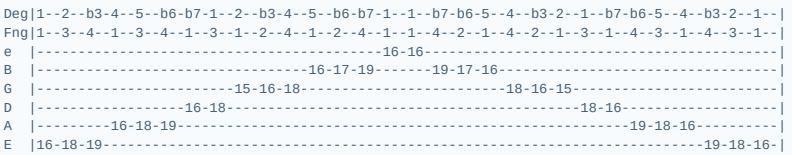
D-start and U-start

12 G# MINOR

ROOT E16 | FRAME 15-19

Notes: G# A# B C# D# E F# | G# A# B C# D# E F# G#

30-NOTE TWO-OCTAVE CELL



Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 22 - Natural Minor Two-Octave Six-String Neck Ladder

Rung 13 and full-ladder consolidation

For the descending pass, return through rungs 12-01 and move -1 fret

13 A MINOR (OCTAVE)

ROOT E17 | FRAME 16-20

Notes: A B C D E F G | A B C D E F G A

30-NOTE TWO-OCTAVE CELL

```
Deg|1--2--b3-4--5--b6-b7-1--2--b3-4--5--b6-b7-1--1--b7-b6-5--4--b3-2--1--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--1--2--4--1--2--4--1--1--1--4--2--1--4--2--1--3--1--4--3--1--4--3--1--|
e  |-----------------------------------------17-17-----------------------------------------|
B  |--------------------------------17-18-20-------20-18-17--------------------------------|
G  |----------------------16-17-19-----------------------19-17-16----------------------|
D  |-----------------17-19-----------------------------------------19-17-----------------|
A  |---------17-19-20-------------------------------------------------------20-19-17----------|
E  |17-19-20-----------------------------------------------------------------------20-19-17-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Full round-trip challenge

1. Start at A natural minor, low-E fret 5.
2. Play all 13 cells without restarting the pick cycle.
3. Reverse after the fret-17 A-minor cell and return through rungs 12-01.
4. Say each key aloud before its low root.
5. Repeat with the opposite starting stroke.

Clean-round standard:

no open strings, wrong accidental, unplanned doubled note, dragged G-string backreach, crossing hesitation, or volume drop in the upper octave.

Advance: complete the route cleanly with both pick starts on two separate days.

TOTAL: 25 cells x 30 notes = 750 picked notes.

After the ladder is clean

Octave accents: accent only the low, middle, and high roots.
Backreach audit: isolate D-to-G; release before the G-string index reach.
Crossing audit: isolate G-to-B and B-to-e; remove landing accents.
Direction: begin selected sessions at fret 17 and descend first.
Random entry: choose any root fret 5-17, name the key, and continue either way.

Practice log

Date	BPM	Start	Clean?	Notes

Advance after two separate clean days, not one unusually good run.